

## Section 1 / Hexadecimal

Hex is only lightly covered here because there is a very good chance you are already familiar with the concept.

The TL;DR is: Hex is base sixteen. Hex is useful because 16 is 2 raised to the power of 4. Each hex digit represents 4 bits. Just two hex digits completely specify the 8 bits in a byte. Hex digits run from 0 through 9 and then A through F.

Example: decimal 255 is hex FF.

### Numbers in Base 10

You already know that each digit in a base 10 number is multiplied by a power of 10. The least significant digit is multiplied by 10 to the 0. The next least significant digit is multiplied to 10 to the 1 and so on.

Floating point numbers in decimal are multiplied by negative powers of 10.

The digits available to us are 0 through 9.

### Numbers in Base 2

Binary is the same thing except with powers of 2. The digits available to us are 0 and 1.

The first (least significant) digit is multiplied by 2 to the 0. The second is multiplied by 2 to the 1 and so on.

The digits of the fractional part of a floating point number are multiplied with negative powers of 2. See Section 2 for more detailed information.

### Number in Base 16 - Hexadecimal

In decimal there are 10 values available for each digit.

In binary there are 2 values available for each digit.

Hex numbers work the same way as binary and decimal except there are 16 values for each digit. We draft the letters A through F to mean the values 10 through 15.

Hex is a handy way of describing binary numbers with much less effort because each hex digit encodes 4 binary digits.

### Octal - Numbers in Base 8

Base 8, or octal, is used far less than hex in CS because base 8 doesn't naturally encode into a powers-of-two centric digital world. Two hex digits perfectly fit

in a byte but two octal digits are 6 bits... too small. Three octal digits are 9 bits... too big.

Goldilocks does not approve of octal.

## Printing Hex

With `printf()` use `%x`.

With `iostream` use the modifier `hex` but note this is sticky. It will remain active until switched back to `dec`.

## The REAL Reason We Love Hex

You can spell stuff with hex.

| Hex Value | Notes                                           |
|-----------|-------------------------------------------------|
| DEADBEEF  | Found frequently in MC68K code especially Amiga |
| 8BADF00D  | Found for similar reasons but on Apple products |
| DEADDEAD  | Associated with the Blue Screen of Death        |
| F003BA11  | Used by undergraduates everywhere               |
| C0EDBABE  | Used by misogynists everywhere                  |
| B16B00B5  | Used by misogynists at Microsoft                |